

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	EggTech-0000000014
Award Program	Egg Tech Prize Phase I & Supplement
Project Title	
Grant Start Date	08/01/2019
Grant End Date	12/31/2022
Total Grant Budget	\$600,000
Total FFAR Budget	\$500,000
Matching Funder(s)	Viscon Hatchery Automation B.V.
Final Report Date	03/31/2023
Project Director/Principal Investigator Information	
Full Name	Dr. Thomas Turpen
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country. *Davis, CA CA-003*
- 1.2. How many new jobs were created because of this grant funding? *1*
- 1.3. How many jobs were maintained because of this grant funding? *4*
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain. *No*
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization. *No*

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Humans rely on protein from chickens to meet the nutritional needs of a growing population. However, only female hens are used for most all food production from egg laying operations. The modern poultry industry faces the problem of how to dispose of billions of male chicks within 24hrs. of hatching. All organisms and their associated microbes release trace gases into the environment in the form of volatile organic compounds (VOCs) and other molecules. Using a variety of statistical methods including machine learning, we and other



researchers have captured and characterized volatiles released from individual eggs as early as 8 days of incubation that comprise a biomolecular signature distinguishing female from male embryos long before chicks hatch. Our active sampling process is submitted for publication and operates by applying moderate vacuum pressure through commercial egg-handling suction cups at incubation temperatures for brief periods of sampling and sorting time. We are in the process of integrating our proprietary miniature chemical sensor microchip, calibrated to detect the discriminatory panel of VOCs, with a custom automated egg handling machine designed in collaboration with a commercial engineering firm. Because the sensor chips can be multiplexed, the equipment is modeled to meet the design specifications of the Egg Tech challenge to sort eggs by sex using an inexpensive, non-invasive, automated and high throughput method. Our goals are now to complete the prototype integration, optimize economic performance, and build and deploy the first commercial hatchery units. Humanely diverting male eggs to other use will be of great societal benefit and further reduce the carbon footprint of poultry production.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The purpose of the Egg Tech Prize is to enable automated sorting of individual chicken eggs by sex without invasive sampling early in incubation. We implemented a moderate vacuum pressure system through commercial egg-handling suction cups to collect volatile organic compounds (VOCs). Our VOC-based and machine learning method could correctly differentiate male from female embryos with more than 80% accuracy. The key findings of our research are described in a publication submitted to the open access journal PLOS One and will therefore be broadly available. This manuscript describes in detail our success in directly sampling the plume of trace gases that are released from chicken eggs after 8 days of incubation with or without the industry practice of SPIDES treatment (Short Period Incubation During Egg Storage). These results provide a solid scientific basis for the future system design of specialized automation equipment capable of high-throughput, in-ovo sexing based on chemical sensor microchips.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:



- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used; *The major outcome of our work is to establish a technical basis for egg sorting as described in the PLOS One manuscript.*
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and *This work has attracted the interest of hatchery automation companies interested in working with us to develop and introduce appropriate equipment for US poultry producers. Eliminating chick-culling is likely to be mandated in the US as technical feasibility is established as is occurring now in the EU.*
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure. *N/A*

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The purpose of the Egg Tech Prize is to enable automated sorting of individual chicken eggs by sex without invasive sampling. Our proposed solution is to collect and analyze volatile organic compounds (VOCs) using a pair of proprietary microchips etched in silicon. The first chip, a micropreconcentrator (uPC) is used to adsorb VOCs onto a matrix. The second chip, is an ion mobility spectrometer (1/2IMS) that can be programmed to identify known chemical entities in complex mixtures. VOCs are sampled by the uPC to increase sensitivity. The matrix is then heated in situ to release these trace gases into the coupled 1/2IMS. This novel hardware and information processing capability must then be integrated into existing robotic egg handling systems. Our project has three major overlapping goals: 1) Biomarker Discovery - YR1, 2) Hardware Calibration (YR1/2) and 3) Design and Demo System (YR2). We have completed 100% of the first goal. We have established an inexpensive lab system to collect VOCs from eggs during incubation and analyze the adsorbed compounds by gas chromatography - mass spectrometry (GC/MS), then determine the sex the chicks directly by DNA analysis using PCR and sex specific primers. This goal was delayed because the GC/MS analytical lab has been intermittently closed by the covid19 pandemic. It was a major effort with many failed experiments to establish this sampling system. The second goal, to calibrate a laboratory system to sample egg derived volatiles directly by IMS, is also 100% complete. Independently, we confirmed using a Photoionization Detector that the total VOC content in a egg-derived plume of trace gas is concentrated enough to be measured in real time without



requiring a uPC. We have made substantial progress (50%) toward the third goal, to Design and Demo the essential components of a high-throughput system. The scaffold of a prototype machine was built and donated to the project by an industry partner as a matching in-kind contribution to a supplemental award from FFAR to continue the research.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

None of the goals of our original proposal have changed. However, for budgetary reasons, we have modified our emphasis on specific objectives with the approval of the sponsor to enhance the prospects of engineering a successful and scalable demo system. The foundation of SensIT technologies are invented, licensed and/or assigned to us by UC Davis through the pioneering work of our co-founder and consultant, Dr. Cristina Davis and their Technology Transfer office, Innovation Access. Our original uPC and 1/2IMS chips were built in a cleanroom on the campus of UC Davis. During our YR1 research and partially because of the FFAR Prize Award, certain commercial foundries have taken an interest in becoming SensIT manufacturing partners. We worked together on design modifications appropriate for the existing tooling at two of these firms and received our first professional runs of our uPC and 1/2IMS chips. This was an important opportunity to accelerate our ability to deliver a full commercial system with scalable components at a known cost, a critical consideration for the Phase II work whether this is funded by the final Egg Tech Prize Competition or by follow-on industry investment. During the period of the award one independent group of researchers in China have published corroborating results using multiple commercial breeds. This is cited in our PLOS One manuscript. Therefore, we used the one commercial breed (Hy-Line 36) that is conveniently available through the UC Davis Avian Facility and focused our efforts more on the sampling conditions and other variables. We added the PID experiments to confirm the concentrations of total VOCs in the plume.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Discovery. *The industry uses a published method to prolong the successful storage of eggs before incubation by incorporating one or two brief periods of increasing temperature before cooling the eggs back to storage temperature and humidity conditions. This is termed Short*



Period Incubation During Egg Storage (SPIDES). We successfully set up a commercially available benchtop egg incubator (Brinsea) with programmable automated egg turning, temperature and humidity controls, and a benchtop refrigeration cabinet to achieve the appropriate conditions to follow a SPIDES protocol without the purchase of the expensive environmental chamber. This procedure was approved by our IACUC. In our discovery work, we used commercially available magnetic solid phase micro-extraction (SPME) fibers that interface with GC/MS (Twister™ - Gerstel). We purchased the type of suction cups used in automated egg handling and connected the cup to a vacuum micropump and mass flow controller. When pressed against the top of an egg this creates a headspace wherein a magnetic Twister™ may be secured within the suction cup with an external magnet. This is not a sealed headspace because the egg shell is gas permeable and in exchange with the chamber air. In a typical experiment, VOCs are collected using a flow rate of 50 ml/min over a 10 min sample period or less. Using principal component analysis (PCA) and partial least squares discriminant analysis (PLS-DA), we have identified compounds ("features") with a Variable Importance in Projection (VIP*) > 1 that contribute to the grouping of male, female and unfertile eggs at high confidence. All of this is described in enough detail for others to reproduce our work in the PLOS One manuscript.

*A VIP score is a measure of a variable's importance in the PLS-DA model. It summarizes the contribution a variable makes to the model. The VIP score of a variable is calculated as a weighted sum of the squared correlations between the PLS-DA components and the original variable. The weights correspond to the percentage variation explained by the PLS-DA component in the model. The number of terms in the sum depends on the number of PLS-DA components found to be significant in distinguishing the classes.

Hardware. The uPC and 1/2IMS chips we received from our commercial foundry partners were evaluated in laboratory test fixtures. The hardware prototype integrates sampling, signal processing, and software analysis. The system collects the VOCs, and passes them through an ionization chamber using either a UV lamp or radioisotope. The ionized gases migrate through an ion mobility spectrometer chip housed in a custom airtight enclosure. The chip then interfaces with custom printed circuit boards (PCBs) to obtain electrical signals from the detector plates at the end of the drift tube. These PCBs were designed to precisely align with the positioning of the inlet and outlet of the chip and improve accuracy and precision of the signals received from the electrodes. The signals obtained are sent via a PXI system (a high-performance modular system that we use for signal processing) to the software stage. The software that interfaces with PXI and other hardware for data processing is LabVIEW. LabVIEW and MATLAB are used to generate dispersion plots and derive measurements of chemical identity, concentration, and standard curves for compounds of interest. The system is tested for leaks with the housing fixture with the 1/2IMS chip and all the tubing, air cylinders, hydrocarbon filters, and mass flow controllers assembled. The PXI-LabVIEW system is connected to the PCBs, and the data measured from the electrodes on the 1/2IMS chip is processed using AIMS (Analyze Ion Mobility Spectrometry) software. Both of the uPC and 1/2IMS chips will need further refinement because the initial foundry runs had flaws with the electrodes and bonding material, respectively. Foundry iterations are to be expected in scale up. We were however, able to bond smaller numbers of chips for prototype development using the UC Davis core facility clean room (CNM2). Using a calibrated PID system, we have established the undiluted plume of egg volatiles contains > 50 ppb of total VOCs.

System. *Creating a low-cost system that is commercially viable is a significant challenge because of both the requirements of throughput and accuracy. We envision a system that is added onto existing egg automation equipment that measures the volatiles directly through the 1/2IMS chip. We have recently obtained Dispersion Plot spectra from eggs using undiluted egg volatiles so the uPC will not likely be necessary (Sionex). To familiarize ourselves with this machinery interface, we have reached out to commercial equipment manufacturers who will also be primarily affected by both the regulatory and technology disruption that will occur with successful resolution of this Prize challenge. The most encouraging discussions have occurred with representatives from Viscon Group who have supplied us with a co-designed framework for prototype sorting. Our Foxhound™ unit containing the 1/2IMS chip is designed to fit over the width of a single egg. In order to achieve the challenge goals, we believe that the ideal system will essentially machine "learn on the job" and thereby improve with a large learning data set that can only be obtained with the throughput achieved in a factory environment. It is critical to emphasize that the simple statistics of the original challenge require a minimal throughput of thousands of measurements per experiment to optimize the system no matter what technology is used (2,401 samples minimum for 2% error, 95% confidence). This was the factual case from the beginning of the challenge description. The 2% error rate is not just an ambitious "stretch" goal for 98% accuracy. It isn't only an animal welfare issue. It is critical to the economics of commercial success. To authentically brand the production system "cull-free" the false negative males must be raised for meat consumption at considerable cost. Commercial success will drive adoption and, ultimately, animal welfare impact at scale. Sustainability, food cost, waste, and animal welfare are all linked. Ours is a systems approach. Together with our automation partner, we have laid out a process line that is based on 3 trays configured in line with 5 rows of multiplexed sensors each, or 150 Foxhound™ units total, to achieve a throughput of 16,200 eggs/hr. The advantage of a multiplex sensor system is precisely that it may operate at the pace of the packing line in real time. In our concept, this is a 100 second cycle. Our conceptual design is fully integrated with conventional process lines for unloading into temperature-controlled cabin for trolley, live embryo detection, gender-ID, and removal of nonviable and male eggs. A final pick and place procedure will consolidate eggs into full trays containing only females that are returned to the incubator. Trays are bar-coded and individual eggs are tracked with existing conventional commercial software with the gender-ID added as a layer in the software database. The eggs experience no physical treatments that are different than normal vacuum handling procedures. Therefore, we do not expect any direct statistically significant impact on hatching rate and have not observed any deleterious effects in the small numbers of chicks hatched.*

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results. *We rely on conventional peer-reviewed publication.*
- 2.8. List key stakeholders that could be served by results of this project. *Hatchery automation manufacturers, poultry producers, general consumers, male chicks.*
- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of



- interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write “nothing to report.” A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit) *Peer-reviewed publication in open access journals (PLOS One). We will coordinate an appropriate news release when the manuscript is published through UC Davis, FFAR, and our own website.*
- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit) *Establishing a reliable system to collect and measure trace egg volatiles with an active sampling method was previously unknown and established by us through this FFAR-sponsored research as an alternative to headspace sampling that requires time periods too long and impractical for commercial use (hours). The pain and the problems solved during a pandemic are too numerous to relive. Using Twisters™ in triplicates and PCR-methods for sexing using chromosomal DNA markers to insure perfect datasets were key. Shortening the sampling time, not letting the eggs cool and determining adequate flow rates are critical.*
- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state “Nothing to Report.” For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit) *Nothing to Report*



- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit) *Nothing to Report*

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award. *PLOS One, open access journal publication in final revisions*
- 3.2. Please provide a list of citations for the information products produced during the project. *This will be provided when available.*
- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation. *Nothing to Report*
- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above. *A link to the manuscript and other news will be posted here: <https://sensit.ventures/>*
- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s). *Nothing to Report*

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit) *Nothing to Report*

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future? *Nothing to Report*

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award. *Contained in PLOS One, open access journal publication.*

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related. *Contained in PLOS One, open access journal publication.*



- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder. *Contained in PLOS One, open access journal publication.*



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

Thomas Turpen

PI Full Name: _____

A handwritten signature in black ink, appearing to read "Tom Turpen", with a long horizontal flourish extending to the right.

PI Signature: _____ Date: _____

March 31, 2023

Dan Hanson

Authorized Signing Official (ASO) Name: _____

A handwritten signature in black ink, appearing to read "Dan Hanson", with a long horizontal flourish extending to the right.

ASO Signature: _____ Date: _____

March 31, 2023